

UNIVERSITY OF BARCELONA
Master Data Science — Time Series Analysis

Predicting Delays on the Barcelona Rodalies R1 Line

A Time-Series Approach

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<https://github.com/ntorqulu/timeseries-analysis/tree/main/PROJECT>

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Abstract

Rodalies de Catalunya recorded only 50 % punctuality in 2025, the worst figure among all Renfe Cercanías networks in Spain, with over 302,000 daily commuters affected. This project focuses on the R1 line (L'Hospitalet de Llobregat – Maçanet-Massanes, 27 stations) and forecasts hourly mean *true operational delay* (`delay_type_1`: actual arrival minus original planned arrival, before any Renfe schedule adjustment) using 78 days of real-time API data (15 March – 31 May 2026), of which 76 days yielded complete service records.

We build a feature dataset integrating four daily data sources: real-time timetable snapshots, train position feeds, hourly weather observations and static network topology. After characterizing the delay distribution and the temporal patterns, we fit and evaluate three families of models seen in class: SARIMA as a univariate seasonal baseline, GARCH(1,1) to model time-varying delay volatility, and SARIMAX, which extends SARIMA with exogenous regressors (cyclic day-of-week encoding, holiday flags and temperature). All models are evaluated on a 72-hour held-out window against a seasonal naive benchmark.

SARIMA(1,0,1)(2,0,1)₂₄ reduces MAE by 13.0 % and RMSE by 18.7 % over the seasonal-naïve baseline. SARIMAX(1,0,1)(1,0,1)₂₄ matches SARIMA in point-forecast accuracy; coefficient analysis reveals that only the day-of-week cosine term is significant for the true operational delay target ($p = 0.027$), while temperature is significant for the passenger-facing delay series ($p < 0.001$). GARCH(1,1) confirms high volatility persistence ($\hat{\alpha} + \hat{\beta} = 0.9485$): a disruption event keeps uncertainty elevated for two to three days.

All results, forecasts, and model parameters are publicly available through an interactive web dashboard at <https://ntorqulu.github.io/timeseries-analysis/>.

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Contribution Statement

Table 1: Percentage contribution per author.

Component	Eleonora Mazzocca	Núria Torquet
Data pipeline	50%	50%
Exploratory data analysis	50%	50%
SARIMA model	50%	50%
GARCH model	50%	50%
SARIMAX model	50%	50%
Report writing	50%	50%

1 Introduction and Motivation

Rodalies de Catalunya is the backbone of the Barcelona metropolitan commuter rail network, carrying over 302,000 passengers on an average weekday (>110 million journeys per year) [1]. For a large fraction of the Catalan working population, the service is not a choice but a necessity.

Yet the network is in a sustained reliability crisis. In 2025 Rodalies recorded only 50 % on-time performance, the worst figure of all Renfe Cercanías networks. And the calendar was marked by a series of high-profile disruptions: strikes in April 2025 [2], a nationwide blackout [3], and a fatal derailment near Gelida in January 2026 caused by Storm Harry [4]. Within our study window, an April 2026 technical fault between L’Hospitalet and Sants forced R1 trains to terminate short: the event is visible as a sharp spike in the training series [5].

If delay magnitude can be predicted with reasonable accuracy, operators can dispatch substitute buses earlier, passenger information systems can issue warnings before commuters reach the platform, and infrastructure managers can correlate predicted spikes with maintenance schedules to prioritize interventions. This motivates a forecasting project rather than a classification one.

2 Objectives

This project uses 76 service days of real-time data to study and model **true operational delay** (`delay_type_1`: actual arrival minus the *original* planned arrival, before any Renfe schedule adjustment) on the R1 line. The main objectives are:

1. Build a clean, reproducible feature dataset integrating four daily data sources, handling schedule-adjustment masking, API gaps, and station normalization.
2. Characterize the distribution, temporal patterns, and propagation of `delay_type_1` across 27 stations.
3. Fit a SARIMA baseline; verify stationarity (ADF, KPSS) and diagnose residuals (ACF, Ljung–Box, ARCH-LM).
4. Model time-varying delay volatility with GARCH(1, 1).
5. Extend to SARIMAX with exogenous regressors; evaluate improvement over the seasonal-naïve benchmark; interpret coefficients.
6. Report MAE and RMSE on a 72-hour held-out test window for all models.

3 Project Diagram

Figure 1 shows the end-to-end pipeline from raw API data to model evaluation. The pipeline has five stages:

1. **Data ingestion.** Five sources are queried daily: three dynamic APIs (`Timetables`, `Trains`, `Weather`) and two static reference tables (`Stations`, `Lines`). Raw snapshots are stored as daily parquet files under `data/dynamic/` and `data/static/`, covering 76 days (15 March – 31 May 2026). The parquet files can be publicly accessed via the following link: https://drive.google.com/drive/folders/1pnnhJDYmW-muiINpIY6cypvvFV0qMx86?usp=drive_link
2. **Feature engineering.** Two Python scripts process the raw files. `build_dataset.py` joins timetables, train positions, and weather; infers journey direction; and computes the two delay variants (`delay_type_1`, `delay_type_2`). `build_weather_dataset.py` aggregates per-minute weather snapshots to an hourly grid and fills gaps via forward fill.
3. **Processed dataset.** The two scripts produce a single `features_YYYYMMDD_HHMMDD.csv` with 35 columns and over 300,000 R1 direction-filtered rows, containing the target delay columns, lagged station delay, hour-of-day, weather covariates, direction, and day type.

4. **Exploratory analysis.** Four notebooks characterise the processed dataset from different angles: data exploration (distributions, missingness), feature EDA (correlations, bottleneck stations), timetable consistency checks, and weather EDA. Their findings directly motivate the modelling choices.
5. **Modelling and evaluation.** Three model families are fitted on the hourly aggregated delay series and evaluated on a **72-hour held-out window** against a seasonal-naïve baseline using MAE and RMSE: SARIMA(1, 0, 1)(2, 0, 1)₂₄ (univariate seasonal baseline with ADF, KPSS, ACF, and Ljung–Box diagnostics), SARIMAX(1, 0, 1)(1, 0, 1)₂₄ extended with day-of-week cyclic encoding, holiday flags, and temperature, and GARCH(1, 1) fitted separately on workday and weekend segments with an ARCH-LM test.

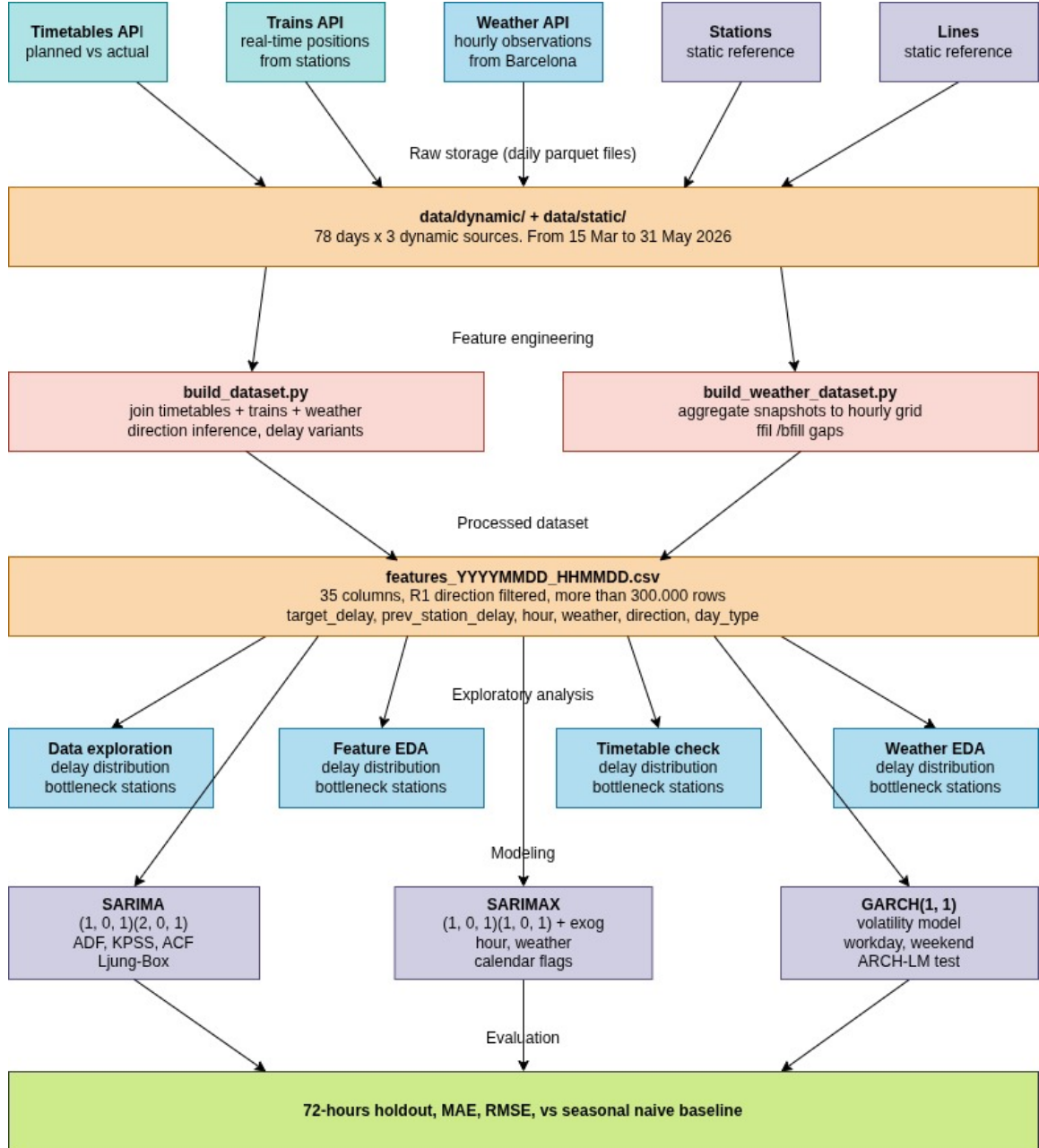


Figure 1: End-to-end data and modelling pipeline: four daily sources are collected, merged into a feature dataset, aggregated to an hourly time series, and passed to three model families evaluated on a held-out test window.

4 Data

4.1 Sources and Collection

Data was collected daily from the Renfe/Rodalies API between 15 March and 31 May 2026, yielding 76 complete service days. Five sources were queried at different frequencies:

Table 2: Data sources.

Source	Frequency	Content
timetables	Per-minute	Planned and actual arrival/departure times per train-stop; both first and last planned arrival are recorded to capture real-time timetable adjustments made by Rodalies.
trains	Per-minute	Real-time train positions and line assignments.
weather	Hourly	Temperature, precipitation, wind speed, cloud cover (Barcelona AEMET API).
stations	Static	Station IDs, names, and coordinates for all 27 R1 stops.
lines	Static	Line identifiers and route metadata.

4.2 Feature Engineering

Two Python scripts process the raw parquet files into our final datasets:

- `build_dataset.py` joins timetable snapshots with train position feeds, infers journey **direction** (northbound vs. southbound) by matching each train against at least two R1 reference stations, and computes the two delay variants. Trains observed at only a single stop cannot have direction inferred and are excluded (11,364 rows, 3.5 %).
- `build_weather_dataset.py` aggregates per-minute weather snapshots to an hourly grid and fills gaps via forward fill. Weather data from raw observations covers 90.9 % of rows; the remainder is carried forward from the most recent available hour.

The output is a single `features_YYYYMMDD_HHMMSS.csv` with 35 columns and 309,959 direction-filtered rows. The key variables are listed in Table 3.

Table 3: Key variables in the processed feature dataset.

Variable	Type	Description
delay_type_1	Continuous	Actual arrival – <i>first</i> planned arrival (min). Main modelling target.
delay_type_2	Continuous	Actual arrival – <i>last</i> planned arrival (min). Passenger-facing delay after Renfe adjustments.
delay_masking_minutes	Continuous	<code>type_1 – type_2</code> : delay absorbed by schedule manipulation.
target_delay	Continuous	Alias for <code>delay_type_1</code> ; the modelling target used in all models.
prev_station_delay	Continuous	<code>delay_type_1</code> at the preceding stop; captures stop-to-stop delay propagation.
hour	Integer	Hour of day (0–23).
day_type	Categorical	<code>workday</code> , <code>weekend</code> , or <code>holiday</code> .
is_weekend	Binary	1 for Saturday/Sunday.
is_holiday	Binary	1 for public holidays.
temperature	Continuous	Hourly temperature (°C).
windspeed	Continuous	Hourly wind speed (km/h).
precipitation	Continuous	Hourly precipitation (mm).
direction	Categorical	Northbound or southbound.

4.3 Dataset Statistics

Table 4: Feature dataset summary (direction-filtered).

Property	Value
Total rows (after direction filter)	309,959
Unique service dates	76
Unique train identifiers	1,046
Unique stations	27
Mean <code>delay_type_1</code>	58.8 min
Mean <code>delay_type_2</code>	7.7 min
Mean <code>delay_masking_minutes</code>	51.1 min
Fraction with <code>type_2</code> >10 min	$\approx 25\%$
Weather from raw observation	90.9%
Rows with missing direction (excluded)	11,364 (3.5%)

5 Exploratory Data Analysis

Before fitting any model, we characterise the processed feature dataset to understand the structure of delays, identify systematic patterns, and motivate the modelling choices that follow. Supplementary figures are collected in Appendix A.

5.1 Delay Definitions and Distribution

The pipeline computes two complementary delay measures for each train-stop observation. `delay_type_1` is clipped to $[-60, 720]$ min and `delay_type_2` to $[-60, 300]$ min to remove API artefacts (spurious ± 1440 min values from canceled services re-entered with incorrect timestamps):

- **Delay Type 1** = actual arrival – *first* planned arrival: the true delay relative to the original timetable, before any real-time adjustment by Renfe. This is the operationally honest measure of how late a train actually is, and the **primary modelling target**. Mean ≈ 58.8 min.
- **Delay Type 2** = actual arrival – *last* planned arrival: the residual delay after Renfe has potentially pushed the scheduled arrival forward to absorb accumulated delay. This is what a passenger sees on the departure board at the time of travel. Mean ≈ 7.7 min.

The gap `delay_masking_minutes` = Type 1 – Type 2 (≈ 51 min on average) quantifies how much real delay Renfe absorbs by silently adjusting the published timetable. The dramatic scale difference between the two series is visible in the hourly plots (Appendix B, Figure 16), which motivates evaluating all models on both definitions as a robustness check.

5.2 Data Quality

Of 321,323 raw rows, 11,364 (3.5%) were removed because direction could not be inferred: direction requires a train to appear at at least two matched R1 reference stations, so trains observed at only a single stop are excluded. The remaining 309,959 rows form `df_model`. Within this filtered dataset, `actual_arrival`, `actual_departure` and `raw_arr_delay` are effectively complete ($< 0.001\%$ missing); missingness is not a modeling concern as can be seen in Figure 6

5.3 Temporal Patterns

The intraday delay profile shows a clear daily seasonality (Figure 8). Delays are lowest in the early morning (05:00–07:00 h, ≈ 1.5 – 2.5 min) when few trains run, rise steeply through the morning peak (06:00–11:00 h), and climb again in the evening (15:00–22:00 h). This U-shaped cycle

repeats across all 76 days and directly motivates the seasonal component $s = 24$ in SARIMA and SARIMAX. At the weekly level, Monday and Tuesday carry the highest average delays (≈ 7.5 – 8 min) while mid-week days are lower (≈ 5.5 min), motivating the day-of-week cyclic regressors and `is_holiday` flag in SARIMAX.

5.4 Spatial Patterns

Delay does not arise uniformly at each stop but accumulates along the route. The worst bottleneck segments are long-distance legs towards the Maçanet-Massanes terminus: Barcelona-Plaça de Catalunya $\rightarrow$ Pineda de Mar adds ≈ 0.17 min of delay per passage on average, which compounds across hundreds of daily services. Correspondingly, terminus-adjacent stations (Malgrat de Mar, Santa Susanna, Pineda de Mar) show the highest mean delays (≈ 8 min), while stations close to L’Hospitalet show the lowest (≈ 6 min). This propagation mechanism is captured by the `prev_station_delay` feature, which correlates at $r = 0.89$ with the target (Appendix A, Figure 14).

5.5 Weather and Feature Correlations

None of the weather variables — temperature, precipitation, wind speed, or cloud cover — show a meaningful linear correlation with delay at the row level (all $|r| < 0.05$). Splitting by rain vs. no rain produces nearly identical delay distributions, and scatter plots confirm the absence of any strong linear signal (Appendix A, Figures 12 and 13). Temperature and wind speed are nonetheless tested as SARIMAX regressors because their row-level noise may mask an aggregate effect; their actual statistical significance is determined empirically by the fitted coefficients in Section 6.2.

5.6 EDA Summary

Four findings directly drive modeling decisions. Strong daily seasonality ($s = 24$) motivates the seasonal components in SARIMA and SARIMAX. The weekday/holiday level shift motivates the `is_holiday` flag and day-of-week cyclic encoding. Stop-to-stop delay propagation ($r = 0.89$) is captured by `prev_station_delay`. Finally, the time-varying variance visible in the hourly series (wider spread at peak hours and sharp spikes during disruption events) motivates the GARCH component, which models this heteroscedasticity directly.

6 Methodology

All three models operate on the hourly mean delay aggregated across all R1 direction-filtered observations. The collection window spans 15 March to 31 May 2026; the last 72 hours form the held-out test window. Overnight hours with no service are filled by linear interpolation.

Table 5: Summary of model specifications.

Model	Purpose	Specification	Regressors
SARIMA	Seasonal univariate baseline	$(1, 0, 1)(2, 0, 1)_{24}$	None
SARIMAX	Extend SARIMA with context	$(1, 0, 1)(1, 0, 1)_{24}$	dow, holiday, temperature
GARCH	Model delay volatility	Constant-mean GARCH(1, 1)	None

Evaluation Baselines

SARIMA and SARIMAX are compared against a **seasonal naive (7-day lag)** baseline: $\hat{y}_t = y_{t-168}$. This is a strong benchmark as it captures the weekly seasonal pattern without fitting any parameters; it is shown as an orange dashed line in the SARIMAX forecast plot (Figure 4). SARIMA is compared against the naive baseline in the metrics table only, as the naive line was not added to its forecast plot (hard to distinguish it from the forecast).

GARCH is not compared against this baseline: its point forecast equals the training mean $\hat{\mu}$ by design. Its value lies in the estimated uncertainty band, not point accuracy.

6.1 SARIMA Baseline

Stationarity and Order Selection

ADF and KPSS tests confirm stationarity for all three delay variants (Table 6), so no differencing is needed ($d = 0$). For **type_1_only** the KPSS statistic (0.460) sits just below the critical value at the 5% level ($p = 0.051$), indicating a borderline result; we proceed with $d = 0$ because the ADF result is unambiguous and the $d = 1$ alternative produces marginally worse accuracy (Table 8).

Table 6: Stationarity test results for the three hourly delay series. All variants pass both tests; **type_1_only** KPSS is borderline ($p = 0.051$) but $d = 0$ is retained.

Target	ADF stat	ADF p	ADF	KPSS stat	KPSS p	KPSS
type_1_only	-5.680	0.000	stationary	0.460	0.051	stationary
type_2_only	-5.748	0.000	stationary	0.125	0.100	stationary
type_1_and_2	-5.960	0.000	stationary	0.324	0.100	stationary

The ACF and PACF plots (Figure 17, Appendix B) guide order selection. The ACF reveals seasonal structure: strong spikes at lags 24, 48, 72, ... confirm $s = 24$ and that a single seasonal AR term ($P = 1$) is insufficient — residual testing with $P = 1$ showed autocorrelation remaining at lag 24, so $P = 2$ is adopted. The seasonal AR coefficient $\hat{\Phi}_1 = 0.984$ is near the seasonal unit root, which is handled by setting **enforce_stationarity=False**; the second seasonal AR term $\hat{\Phi}_2 = 0.018$ is not individually significant ($p = 0.318$) but the $P = 2$ specification as a whole improves residual diagnostics. The non-seasonal PACF cuts off after lag 1, supporting $p = 1$.

Model Specification

$$(1 - \phi_1 B)(1 - \Phi_1 B^{24} - \Phi_2 B^{48}) y_t = (1 + \theta_1 B)(1 + \Theta_1 B^{24}) \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2) \quad (1)$$

where B is the backshift operator, ϕ_1 and θ_1 are the non-seasonal AR and MA coefficients, and Φ_1, Φ_2, Θ_1 their seasonal counterparts at period $s = 24$. For **type_1_only** the fitted coefficients are $\hat{\phi}_1 = 0.669$, $\hat{\Phi}_1 = 0.984$, $\hat{\Phi}_2 = 0.018$, $\hat{\Theta}_1 = -0.937$, $\hat{\sigma}^2 = 102.7$. Note that $\hat{\phi}_1$ is significant ($p < 0.001$) but $\hat{\theta}_1 = 0.060$ is not ($p = 0.076$), suggesting the non-seasonal MA term adds little for this target.

Forecast and Residual Diagnostics

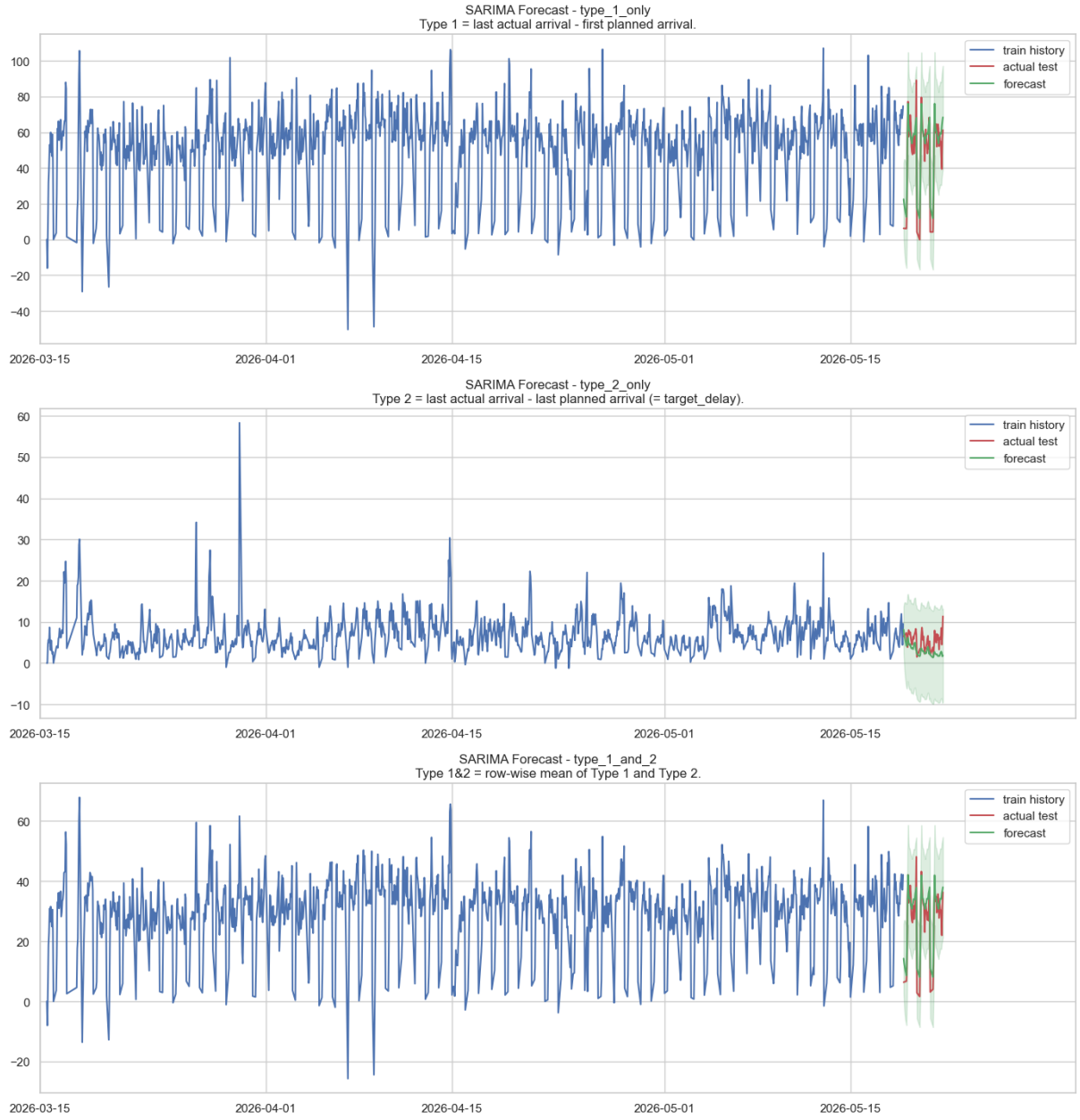


Figure 2: SARIMA(1, 0, 1)(2, 0, 1)₂₄ forecast on the 72-hour test window for all three delay variants (blue: training history; red: actual test; green: forecast; shaded: 95 % CI). For **type_1_only** (top) the CI is wide and the forecast converges to the training mean, reflecting the high variance of the unmasked delay series. For **type_2_only** (middle) the intraday pattern is tracked more closely.

Ljung–Box residual tests reveal an important difference between targets (Table 7). For **type_2_only** no significant autocorrelation remains at lag 24 ($p = 0.085$) or lag 168 ($p = 0.494$) — the model is well-specified. For **type_1_only** significant autocorrelation persists at both lags ($p < 0.001$), indicating that the unmasked delay series contains structure that pure seasonal ARMA cannot capture. The $D = 1$ seasonal differencing alternative is strictly worse: Ljung–Box $p = 0.000$ at both lags. This residual structure in **type_1_only** directly motivates SARIMAX, where exogenous regressors capture the calendar-driven level shifts responsible for the remaining autocorrelation.

Table 7: Ljung–Box test on SARIMA residuals. `type_2_only` passes at both lags; `type_1_only` fails, motivating SARIMAX.

Target	Lag	LB stat	<i>p</i> -value	Conclusion
<code>type_1_only</code>	24	68.93	< 0.001	autocorr remains
<code>type_1_only</code>	168	230.29	0.001	autocorr remains
<code>type_2_only</code>	24	33.99	0.085	white noise ✓
<code>type_2_only</code>	168	167.63	0.494	white noise ✓
<code>type_1_and_2</code>	24	60.53	< 0.001	autocorr remains
<code>type_1_and_2</code>	168	213.05	0.011	autocorr remains
<code>type_1_only</code> $D=1$	24	125.49	< 0.001	autocorr remains
<code>type_1_only</code> $D=1$	168	273.19	< 0.001	autocorr remains

Table 8: SARIMA forecast accuracy on the 72-hour test window vs. seasonal-naïve baseline. $D=1$ is tested as an alternative for `type_1_only` and performs marginally worse.

Series	Naïve MAE	Naïve RMSE	SARIMA MAE	SARIMA RMSE	MAE Δ	RMSE Δ
<code>type_1_only</code>	10.800	16.458	9.397	13.388	+13.0 %	+18.7 %
<code>type_2_only</code>	2.236	3.031	2.335	2.934	−4.4 %	+3.2 %
<code>type_1_and_2</code>	6.044	9.246	5.247	7.303	+13.2 %	+21.0 %
<code>type_1_only</code> $D=1$	10.800	16.458	9.420	13.460	+12.8 %	+18.2 %

SARIMA beats the seasonal-naïve baseline by 13 % MAE and 18.7 % RMSE on `type_1_only`. On `type_2_only` SARIMA is marginally worse than naïve on MAE (−4.4 %), reflecting that Renfe’s schedule adjustments compress that series to such a narrow range (≈ 2 –3 min variation) that simple pattern repetition is already a strong benchmark.

6.2 SARIMAX with Exogenous Regressors

Motivation and Specification

The Ljung–Box test on SARIMA residuals showed significant autocorrelation remaining at lags 24 and 48 for `type_1_only`. SARIMAX extends SARIMA with a linear regression on exogenous inputs $\mathbf{x}_t$ to absorb the calendar-driven level shifts responsible for that residual structure:

$$(1 - \phi_1 B)(1 - \Phi_1 B^{24})(y_t - \beta^\top \mathbf{x}_t) = (1 + \theta_1 B)(1 + \Theta_1 B^{24})\varepsilon_t \quad (2)$$

The final regressor set is `dow_sin`, `dow_cos` (cyclic day-of-week encoding, period 7), `is_holiday`, and `temperature`. Variables `hour_sin/cos` were excluded due to collinearity with the seasonal AR/MA terms; `windspeed` was excluded ($p = 0.750$ in a preliminary fit). Cyclic encoding maps each day of the week to a point on the unit circle, $(\sin(2\pi d/7), \cos(2\pi d/7))$, avoiding the artificial discontinuity of raw integer encoding. A prior OLS screen on the hourly aggregate series (Table 9) confirmed that temperature has the strongest aggregate correlation with `type_1_only` ($r = 0.287$), justifying its inclusion despite near-zero row-level correlation. The seasonal order is reduced to $(1, 0, 1)_{24}$ from the SARIMA $(2, 0, 1)_{24}$ because the exogenous regressors are expected to absorb part of the two-day seasonal memory that required $P = 2$ in the univariate model.

Table 9: OLS pre-screen: Pearson correlation of exogenous features with hourly aggregate delay. Temperature shows the strongest aggregate signal for `type_1_only`.

Feature	r with <code>type_1</code>	r with <code>type_2</code>
temperature	0.287	0.094
dow_cos	0.030	0.152
dow_sin	−0.049	0.048
is_holiday	−0.041	−0.097

Fitted Coefficients

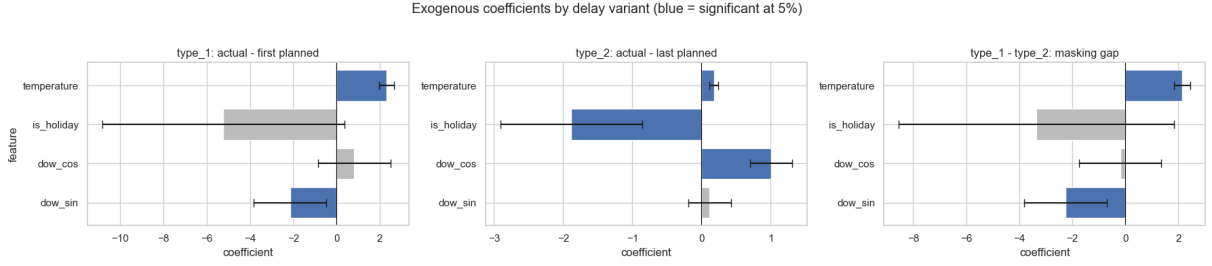


Figure 3: SARIMAX exogenous coefficients with 95 % CI for all three delay variants (blue = significant at $p < 0.05$; grey = not significant). For `type_1_only` (left): only `dow_cos` is significant ($\hat{\beta} = +1.97$ min, $p = 0.027$); `temperature` is not significant in the full SARIMAX model ($\hat{\beta} = +0.052$ min, $p = 0.860$) despite its aggregate OLS correlation, because the seasonal AR/MA structure absorbs most of the temperature-driven variation once the full time series context is included. For `type_2_only` (centre): `dow_cos` (+1.01 min, $p < 0.001$), `temperature` (+0.36 min, $p < 0.001$), and `is_holiday` (−1.48 min, $p < 0.001$) are all significant. The contrast between targets is a key finding: temperature reaches significance for the passenger-facing delay (Type 2) but not for the true operational delay (Type 1), suggesting that Renfe’s schedule adjustments are themselves sensitive to weather conditions.

Forecast

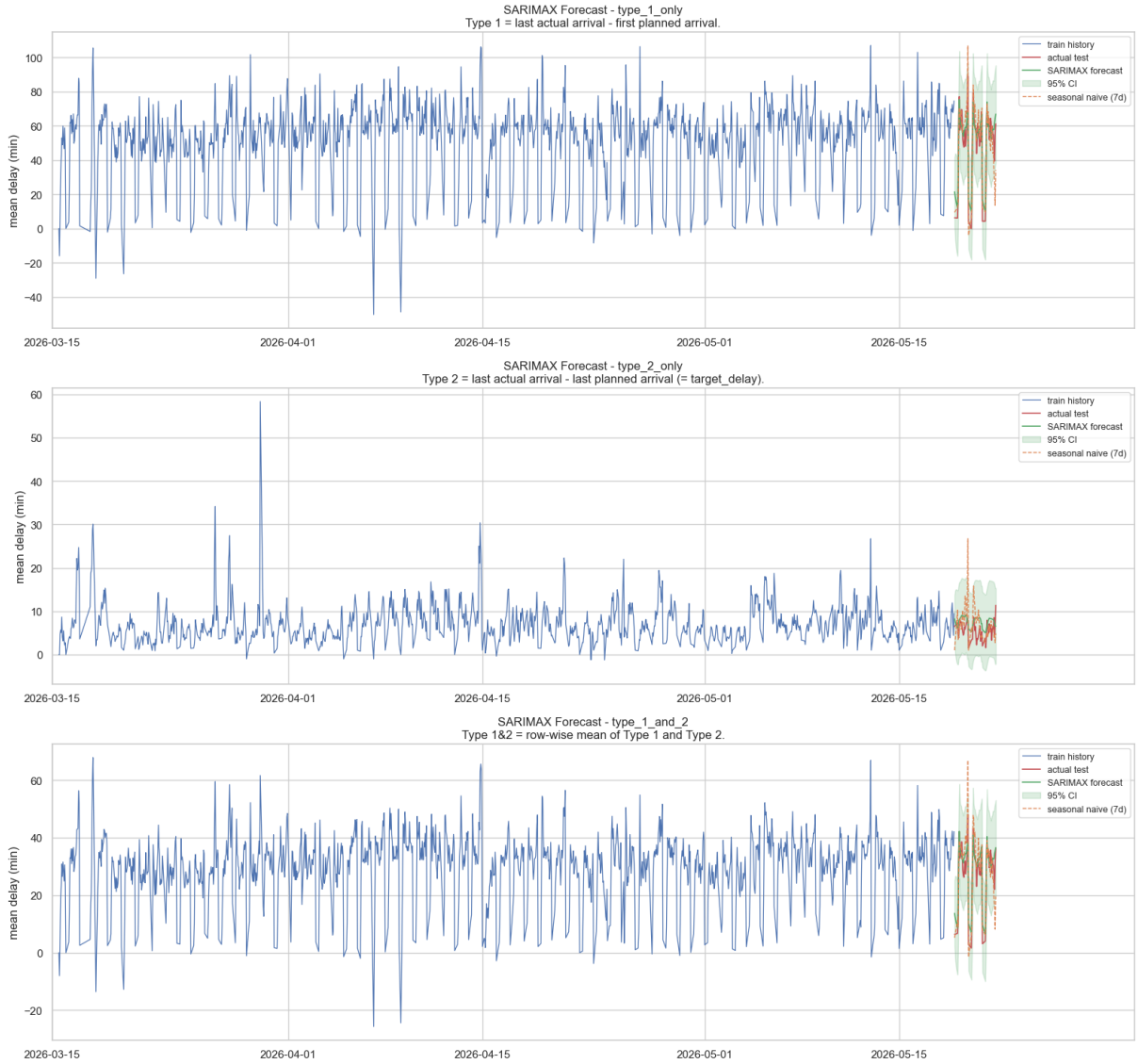


Figure 4: SARIMAX(1,0,1)(1,0,1)₂₄ forecast vs. seasonal-naïve baseline (orange dashed) for all three delay variants (blue: training; red: actual test; green: SARIMAX forecast; shaded: 95 % CI). The split date is 2026-05-28. For **type_1_only** (top) SARIMAX tracks the test level well and clearly outperforms the naïve baseline. For **type_2_only** (middle) both SARIMAX and the naïve baseline produce similar forecasts; the test window falls in a calm period.

Residual Diagnostics

Ljung–Box tests on SARIMAX residuals for **type_1_only** show that autocorrelation remains at lags 24 ($p = 0.006$) and 48 ($p < 0.001$), but the weekly structure is resolved at lag 168 ($p = 0.055$, borderline). Adding exogenous regressors therefore removes the weekly pattern but leaves some residual sub-daily structure. The ARCH-LM test on the residuals gives $p < 0.001$, confirming significant volatility clustering: the residual variance is not constant but spikes during disruption events and decays gradually. This directly motivates the GARCH model in Section 6.3, which models this time-varying uncertainty explicitly.

Table 10: SARIMAX forecast accuracy on the 72-hour test window. **type_1_only** improves substantially over naïve; **type_2_only** is worse than naïve because the exogenous regressors add noise to an already smooth, low-variance series.

Series	Naïve MAE	Naïve RMSE	SARIMAX MAE	SARIMAX RMSE	MAE Δ	RMSE Δ
type_1_only	10.800	16.458	9.392	13.579	+13.0 %	+17.5 %
type_2_only	2.236	3.031	2.566	3.161	−14.8 %	−4.3 %
type_1_and_2	6.044	9.246	5.503	7.662	+9.0 %	+17.1 %

6.3 GARCH for Delay Volatility

Motivation and Specification

The ARCH-LM test on SARIMAX residuals ($p < 0.001$) confirms that residual variance is not constant: disruption periods show both higher mean delay and higher variance, then calm periods follow. This **volatility clustering** is not captured by SARIMA or SARIMAX, which model only the conditional mean. GARCH(1, 1) models the time-varying conditional variance σ_t^2 directly:

$$y_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1) \quad (3)$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (4)$$

The constant mean μ is intentional: GARCH answers “*how uncertain will the delay be?*” rather than “*what will the delay be?*”. α (ARCH term) captures how sharply a new shock inflates uncertainty; β (GARCH term) captures how slowly that elevated uncertainty decays. The stationarity condition $\alpha + \beta < 1$ must hold.

Why Segment by Day Type?

A constant-mean GARCH has no mechanism to handle the workday/weekend level difference ($\hat{\mu} \approx 54$ min workdays vs. ≈ 52 min weekends for **type_1_only**). Fitting on the mixed series would produce a single $\hat{\mu}$ averaging two distinct regimes, yielding a conditional variance estimate that misrepresents both. The model is therefore fitted on three segments: **all_week** (primary comparison target, 1,752 training hours), **workday** (1,138 stacked service hours), and **weekend** (397 stacked service hours).

Fitted Parameters

Table 11 reports the fitted parameters for all nine target–segment combinations. All **type_1_only** fits are well-behaved with high persistence ($\hat{\alpha} + \hat{\beta} \approx 0.948\text{--}0.950$): once the R1 line enters a high-uncertainty regime, elevated uncertainty persists for 48–72 hours even after the mean delay returns to baseline. The ARCH term dominates ($\hat{\alpha} \approx 0.70\text{--}0.86$), indicating that a new shock inflates the uncertainty band rapidly.

For **type_2_only**, the **workday** and **weekend** segments degenerate to ARCH(1) ($\hat{\beta} = 0$, $p = 1.000$): the schedule-adjusted series is too smooth for the optimizer to separately identify the persistence term. Confidence intervals for those two panels are unreliable and are flagged in the forecast plot. The **all_week** segment converges ($\hat{\alpha} + \hat{\beta} = 1.000$, borderline non-stationary) and its CI should also be interpreted cautiously.

Table 11: Fitted GARCH(1,1) parameters for all nine target–segment combinations. Degenerate fits ($\hat{\beta} = 0$) are marked with †; their confidence intervals are unreliable.

Target	Segment	$\hat{\mu}$	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	Status
type_1_only	all_week	53.64	93.16	0.790	0.159	0.949	OK
type_1_only	workday	53.96	108.84	0.858	0.089	0.947	OK
type_1_only	weekend	52.43	79.93	0.695	0.255	0.950	OK
type_2_only	all_week	5.14	1.83	0.662	0.339	1.000	borderline
type_2_only	workday	5.65	3.53	0.937	0.000	0.937	† ARCH(1)
type_2_only	weekend	5.67	5.03	0.664	0.000	0.664	† ARCH(1)
type_1_and_2	all_week	29.89	32.31	0.729	0.163	0.892	OK
type_1_and_2	workday	30.27	36.28	0.813	0.093	0.906	OK
type_1_and_2	weekend	28.74	23.76	0.600	0.309	0.909	OK

Forecast

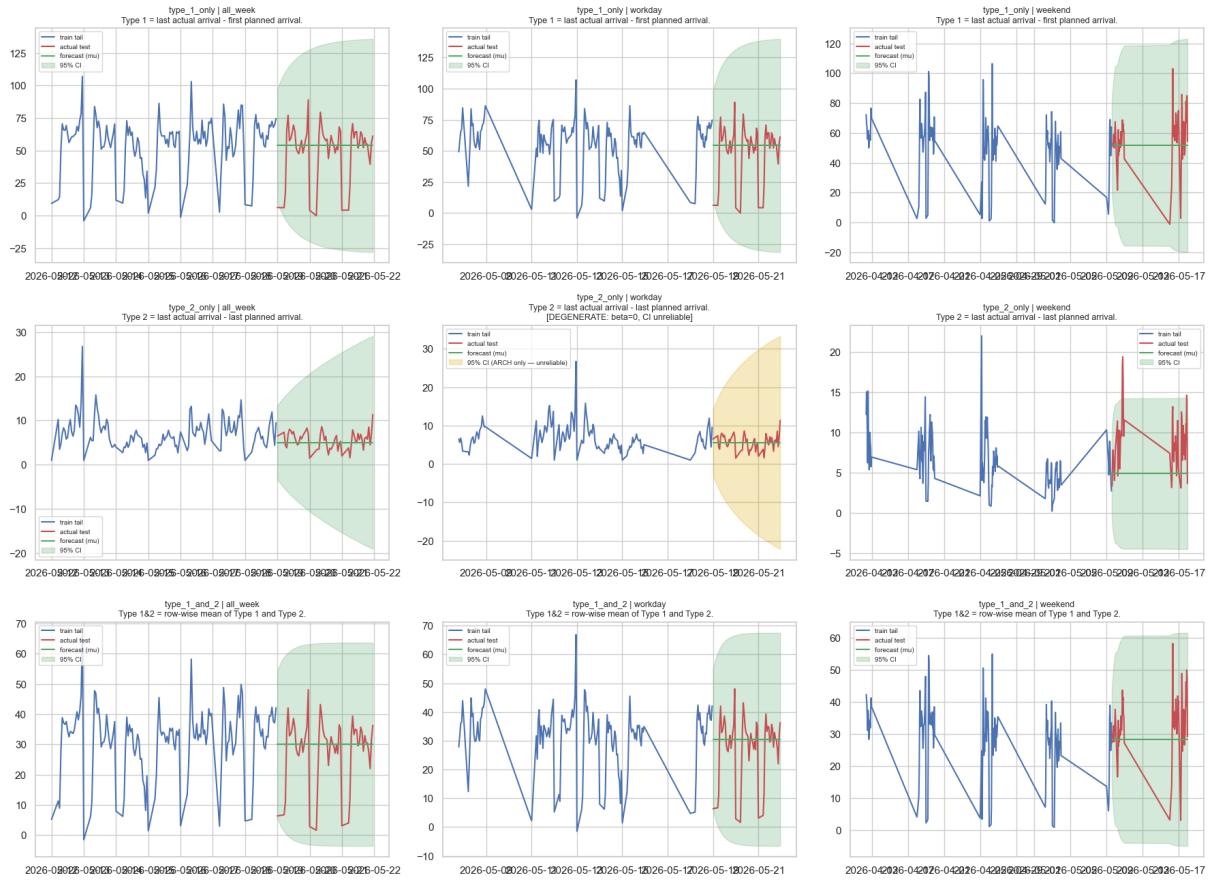


Figure 5: GARCH(1,1) forecasts for all nine target–segment combinations (blue: training tail; red: actual test; green: flat mean $\hat{\mu}$; green shading: 95% CI; yellow shading: degenerate CI, unreliable). The point forecast is flat by design — the operationally relevant output is the **band width**. For `type_1_only | all_week` (top left) the CI correctly captures the volatile test period. The `type_2_only | workday` and `weekend` panels (middle row, centre and right) are marked DEGENERATE because $\hat{\beta} = 0$: the GARCH(1,1) collapsed to ARCH(1) and the CI is unreliable. Conditional volatility plots are in Appendix D.

7 Results

Table 12 summarises point-forecast accuracy on the 72-hour held-out window against the seasonal-naïve benchmark for the primary target `type_1_only` (true operational delay).

Table 12: Forecast accuracy on the 72-hour test window (**type_1_only**, true operational delay). Improvement is relative to the seasonal-naïve baseline. GARCH is a volatility model; its flat-mean MAE/RMSE are shown for completeness only.

Model	MAE (min)	RMSE (min)	MAE impr.	RMSE impr.
Seasonal naïve (168-h lag)	10.800	16.458	—	—
SARIMA(1, 0, 1)(2, 0, 1) ₂₄	9.397	13.388	+13.0 %	+18.7 %
SARIMAX(1, 0, 1)(1, 0, 1) ₂₄	9.392	13.579	+13.0 %	+17.5 %
GARCH(1, 1) flat mean	18.360	23.886	<i>volatility model only</i>	

Key findings from the model evaluations:

- **SARIMA and SARIMAX both beat the seasonal-naïve baseline** on **type_1_only**: MAE drops from 10.80 min to ≈ 9.39 min (+13.0 %) and RMSE from 16.46 min to 13.4–13.6 min (+17.5–18.7 %). The two models perform near-identically, confirming that the exogenous regressors add marginal point-forecast accuracy beyond the ARMA structure. However, SARIMA Ljung–Box residuals fail at lags 24 and 48 ($p < 0.01$), indicating unmodelled sub-daily structure; SARIMAX resolves the weekly pattern (lag 168 $p = 0.055$) but residual autocorrelation at shorter lags remains.
- **On type_2_only** (passenger-facing delay) both SARIMA (−4.4 % MAE) and SARIMAX (−14.8 % MAE) perform *worse* than the naïve baseline. Renfe’s schedule adjustments compress that series to such a narrow range (≈ 2 –3 min variation) that simple pattern repetition is already a strong benchmark, leaving little for the models to improve on.
- **SARIMAX coefficient interpretation:** for **type_1_only** only **dow_cos** is statistically significant ($\hat{\beta} = +1.97$ min, $p = 0.027$); temperature is not significant ($p = 0.860$). For **type_2_only**, temperature (+0.36 min/°C, $p < 0.001$) and **dow_cos** (+1.01 min, $p < 0.001$) are both significant. The contrast suggests that temperature drives true operational delay but Renfe absorbs most of this effect through schedule adjustments before it reaches the passenger-facing metric.
- **GARCH(1, 1)** is not a point-forecast competitor: its flat mean produces MAE 18.36 min by design. Its value lies in the uncertainty band. All **type_1_only** segments show high persistence ($\hat{\alpha} + \hat{\beta} \approx 0.947$ –0.950): a disruption event keeps the confidence interval elevated for 48–72 hours even after the mean delay returns to baseline. For **type_2_only** **workday** and **weekend** the fit degenerates to ARCH(1) ($\hat{\beta} = 0$); confidence intervals for those panels are unreliable.
- **The April 2026 disruption spike** appears as a large residual in all models. No covariate-based model can anticipate an unannounced technical failure, which sets a practical ceiling on forecast accuracy with the current feature set.

8 Conclusion and Future Work

8.1 Have We Achieved Our Objectives?

All six objectives stated in Section 2 were met:

1. A reproducible four-source feature dataset was built; the delay-masking phenomenon was quantified (≈ 51 min per observation absorbed by Renfe schedule adjustments); 96.4 % of rows were retained after direction filtering.
2. EDA revealed a clear $s = 24$ intraday seasonal structure, a workday/weekend level shift, stop-to-stop delay propagation ($r = 0.89$ for **prev_station_delay**), and time-varying variance, all confirmed by ACF/PACF analysis and heteroscedasticity tests.
3. ADF and KPSS confirmed stationarity ($d = 0$, with a borderline KPSS result for **type_1_only**, $p = 0.051$); SARIMA(1, 0, 1)(2, 0, 1)₂₄ reduced MAE by 13.0 % and RMSE by

- 18.7% over the seasonal-naïve baseline. Ljung–Box tests revealed residual autocorrelation at lags 24 and 48 for `type_1_only`, motivating SARIMAX.
4. GARCH(1, 1) confirmed very high volatility persistence for `type_1_only` ($\hat{\alpha} + \hat{\beta} = 0.9485$ all_week), an operationally actionable uncertainty quantification. Two segments (`type_2_only` `workday` and `weekend`) degenerated to ARCH(1), highlighting the limits of GARCH on short, smooth series.
 5. SARIMAX matched SARIMA in point-forecast accuracy. Only `dow_cos` is significant for `type_1_only` ($p = 0.027$); temperature is significant for `type_2_only` ($p < 0.001$) but not for `type_1_only` ($p = 0.860$), revealing that Renfe’s schedule adjustments absorb the temperature signal before it reaches the passenger-facing metric.
 6. MAE and RMSE were reported for all models and all three delay variants on the 72-hour held-out window.

8.2 Future Work

Model extensions. The most natural next step is a **LSTM** (Long Short-Term Memory) network, which can learn non-linear delay dynamics and multi-step patterns beyond the reach of linear ARMA. A **Reservoir Computing Network** (Echo State Network) would be a lighter-weight alternative for capturing long-range dependencies without full backpropagation. An **ARIMAX with a GARCH error process** would directly combine mean and variance modelling in a single framework, addressing the residual ARCH effects that SARIMAX only partially removes.

Data extensions. Only 76 days of data limits seasonal model reliability; a full year would allow $s = 168$ (weekly) seasonality to be estimated reliably. External event indicators (planned engineering works, weather amber/red alerts) could be added as intervention variables to absorb the large disruption spikes that currently inflate residuals.

Operational deployment. The GARCH volatility output could be surfaced in passenger apps as a “delay uncertainty index”, providing actionable information before commuters reach the platform. The interactive dashboard at <https://ntorqulu.github.io/timeseries-analysis/> already provides a foundation for this.

A EDA Supplementary Figures

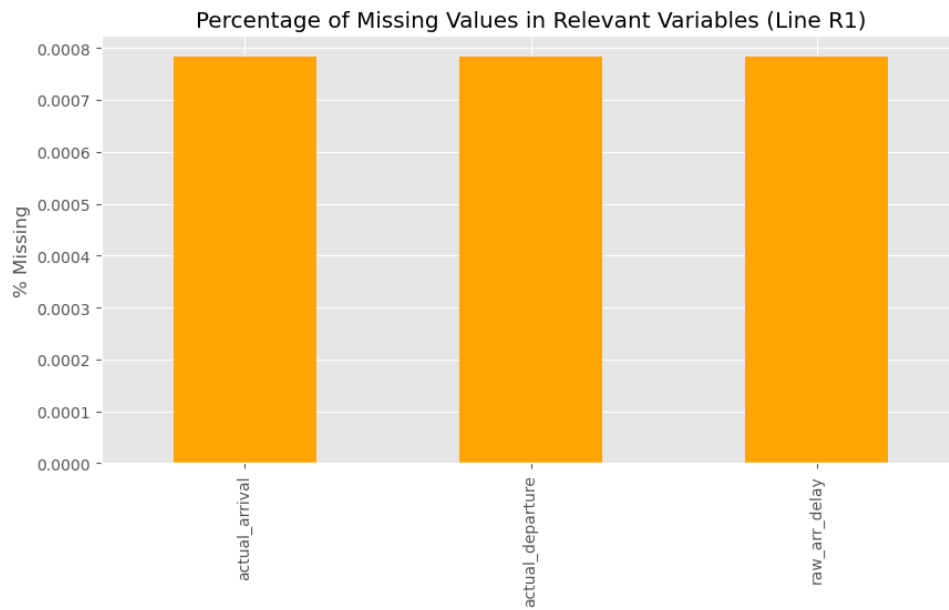


Figure 6: Percentage of missing values in `actual_arrival`, `actual_departure`, and `raw_arr_delay` after direction filtering. All three are below 0.001 % missing.

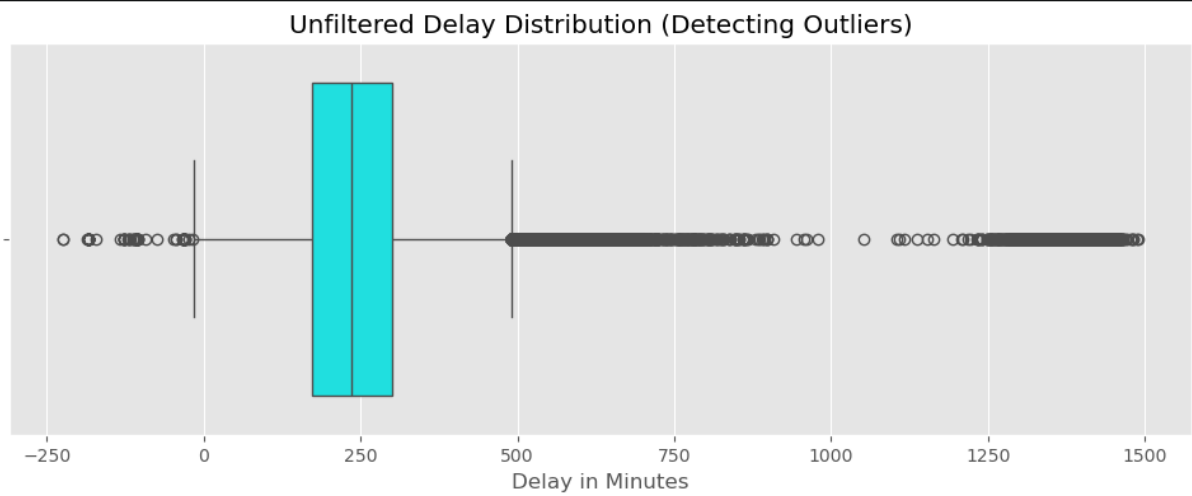


Figure 7: Unfiltered distribution of `target_delay` (box plot, clipped to $[-60, 300]$ min). Median 4.0 min, mean 7.6 min, std 10.76 min; 23.9 % of observations exceed 10 min.

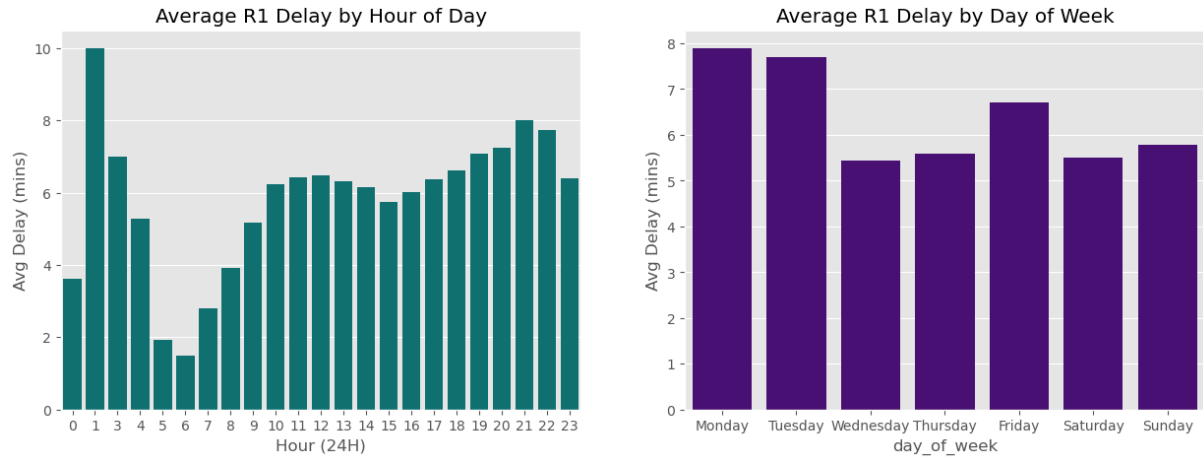


Figure 8: Average delay by hour of day (left) and day of week (right). The intraday U-shaped profile and Monday/Tuesday peaks motivate the seasonal component and day-of-week cyclic regressors.

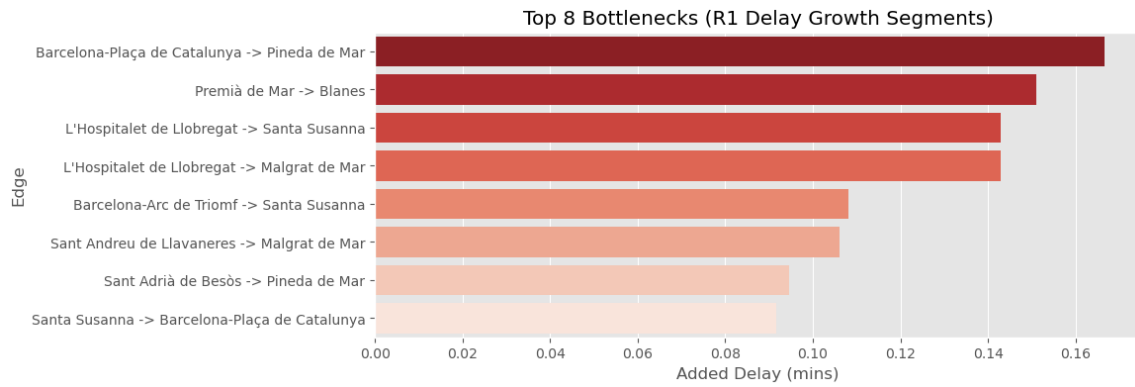


Figure 9: Top 8 bottleneck segments by mean added delay (min). Long-distance legs toward Maçanet-Massanes dominate, confirming delay propagation along the route.

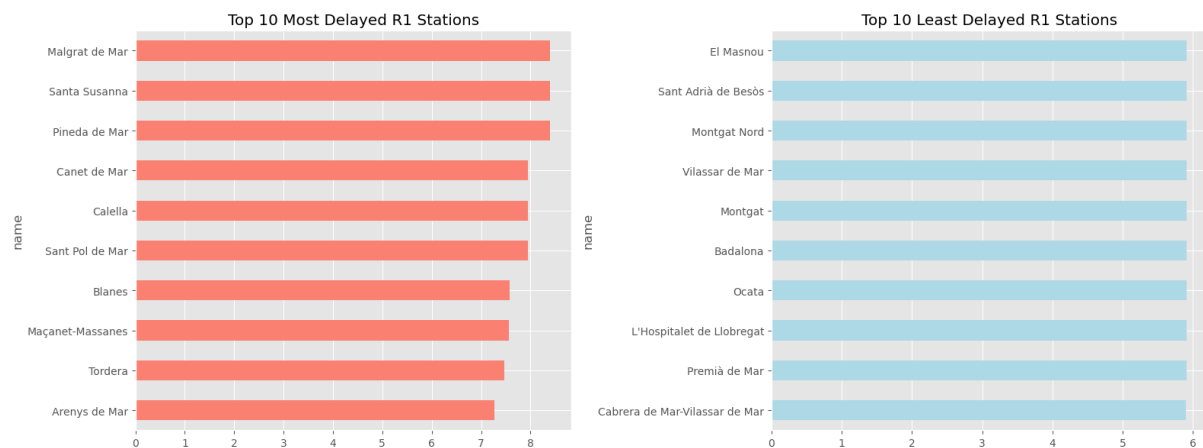


Figure 10: Top 10 most delayed (left) and least delayed (right) stations by mean `target_delay`.

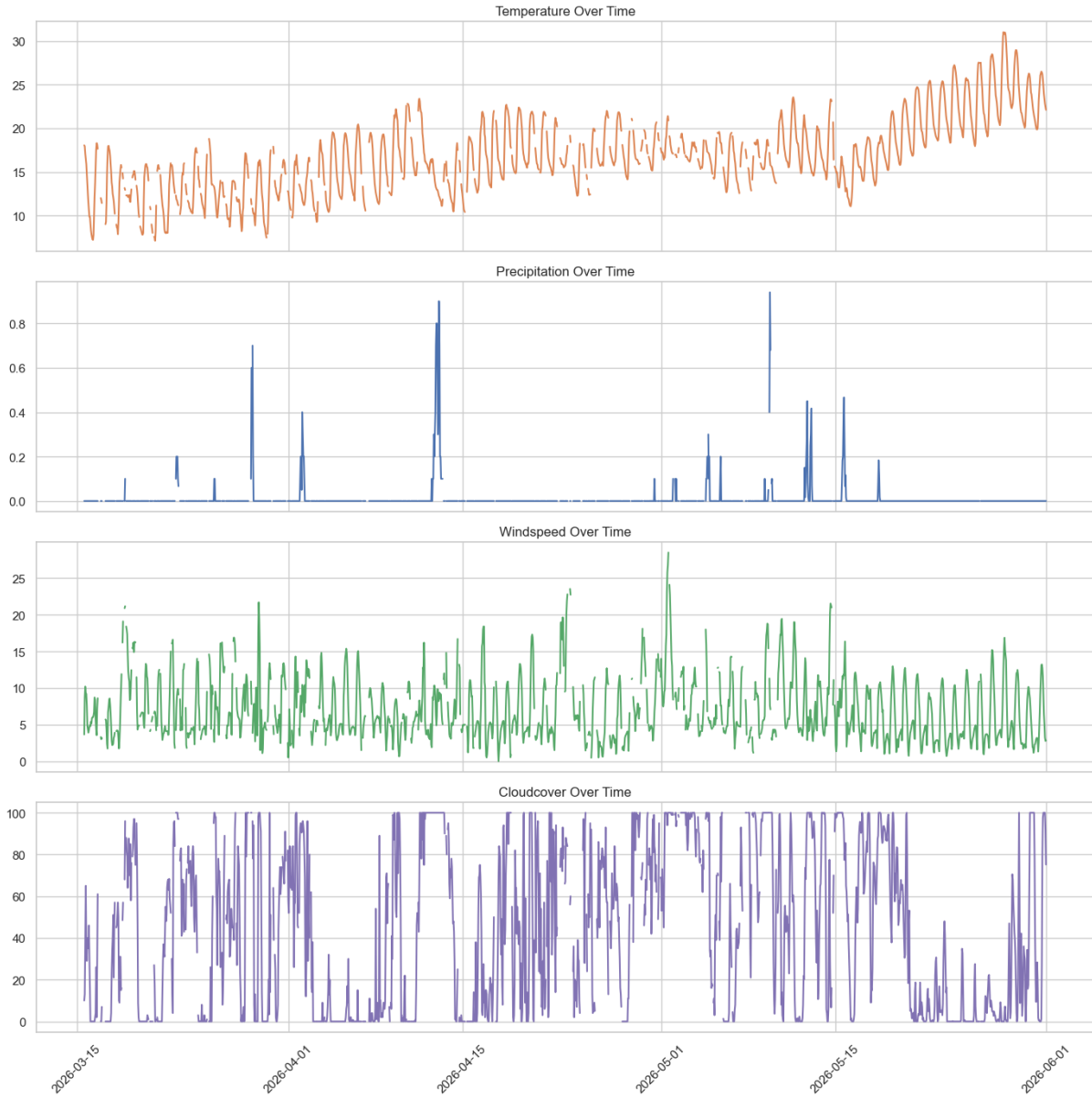


Figure 11: Evolution of temperature, precipitation, wind speed, and cloud cover over the 76-day collection window. Temperature shows a clear upward trend from mid-March to late May.

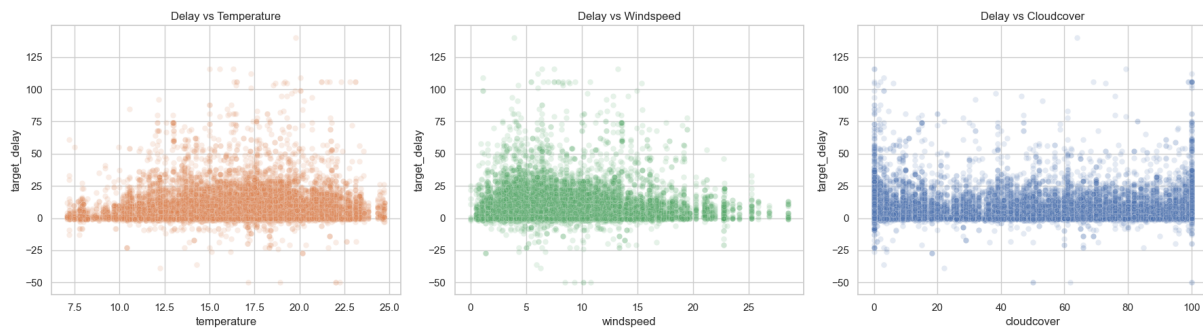


Figure 12: Scatter plots of `target_delay` vs temperature, wind speed, and cloud cover. No strong linear relationship is visible at the row level for any variable.

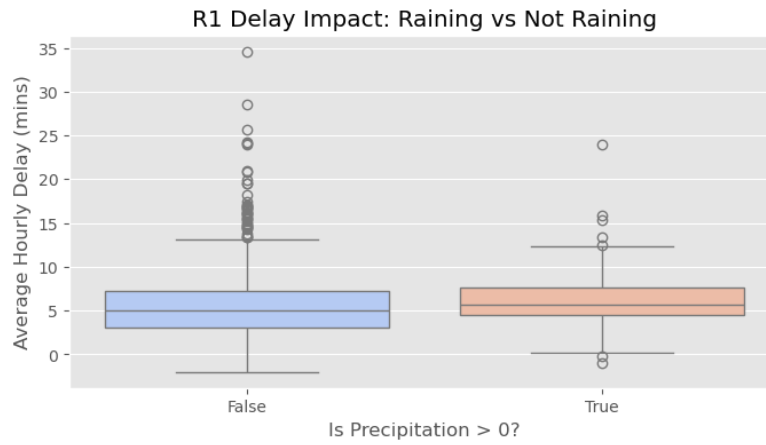


Figure 13: Delay distribution split by precipitation (True/False). Distributions are nearly identical, confirming no meaningful rain effect on delay.

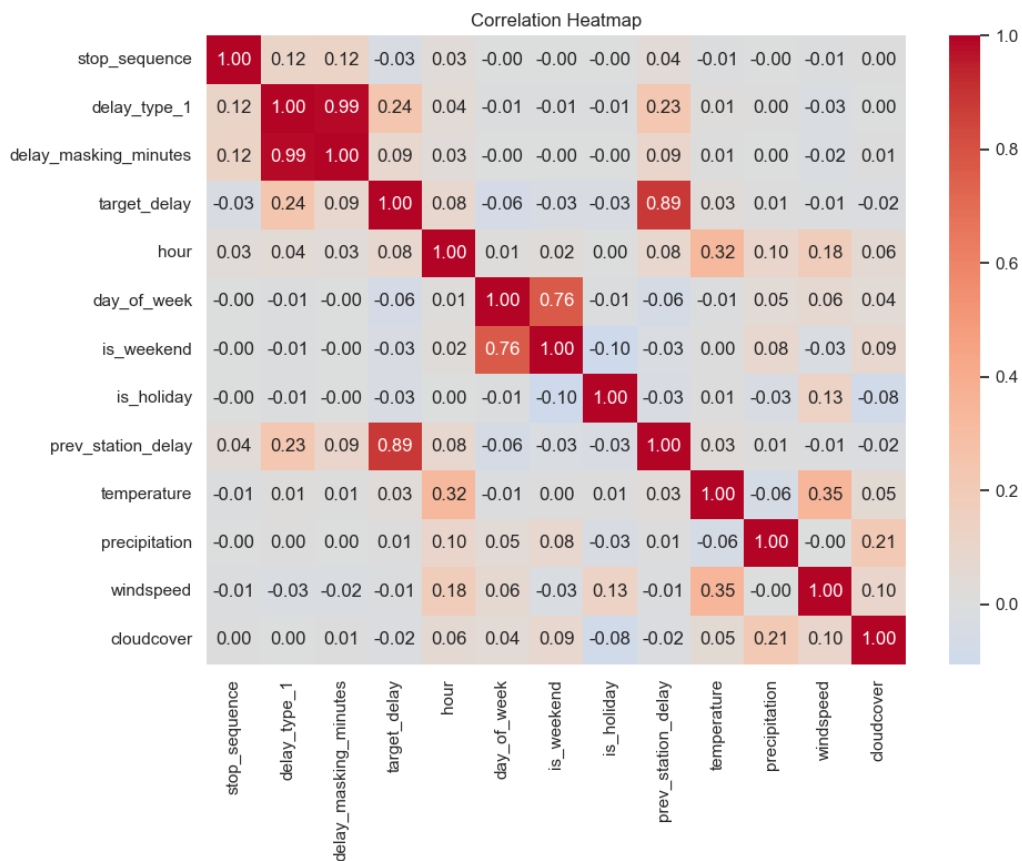


Figure 14: Pearson correlation heatmap. Key values: `prev_station_delay` vs `target_delay` $r = 0.89$; all weather variables $|r| < 0.05$.

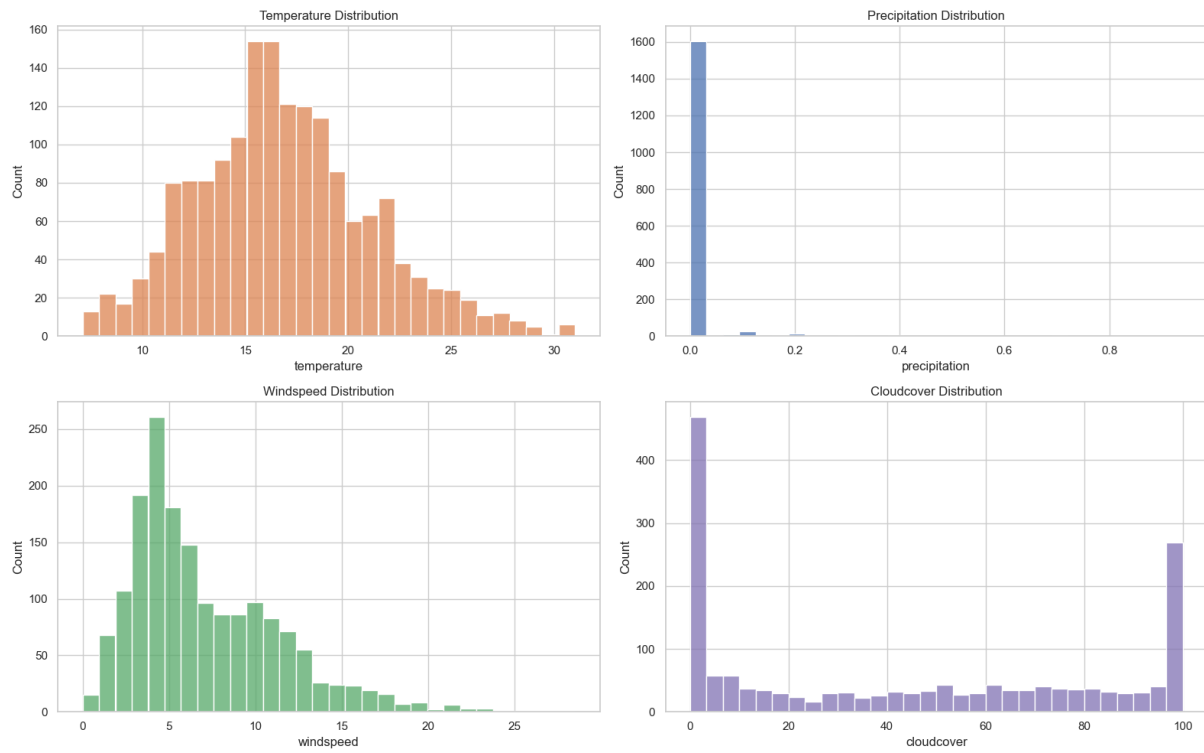


Figure 15: Distributions of the four hourly weather variables across the collection period.

B SARIMA Supplementary Figures



Figure 16: Hourly mean delay series for all three target variants across the full training window. Type 1 fluctuates around 50–75 min; Type 2 is compressed to 0–15 min by Renfe schedule adjustments.

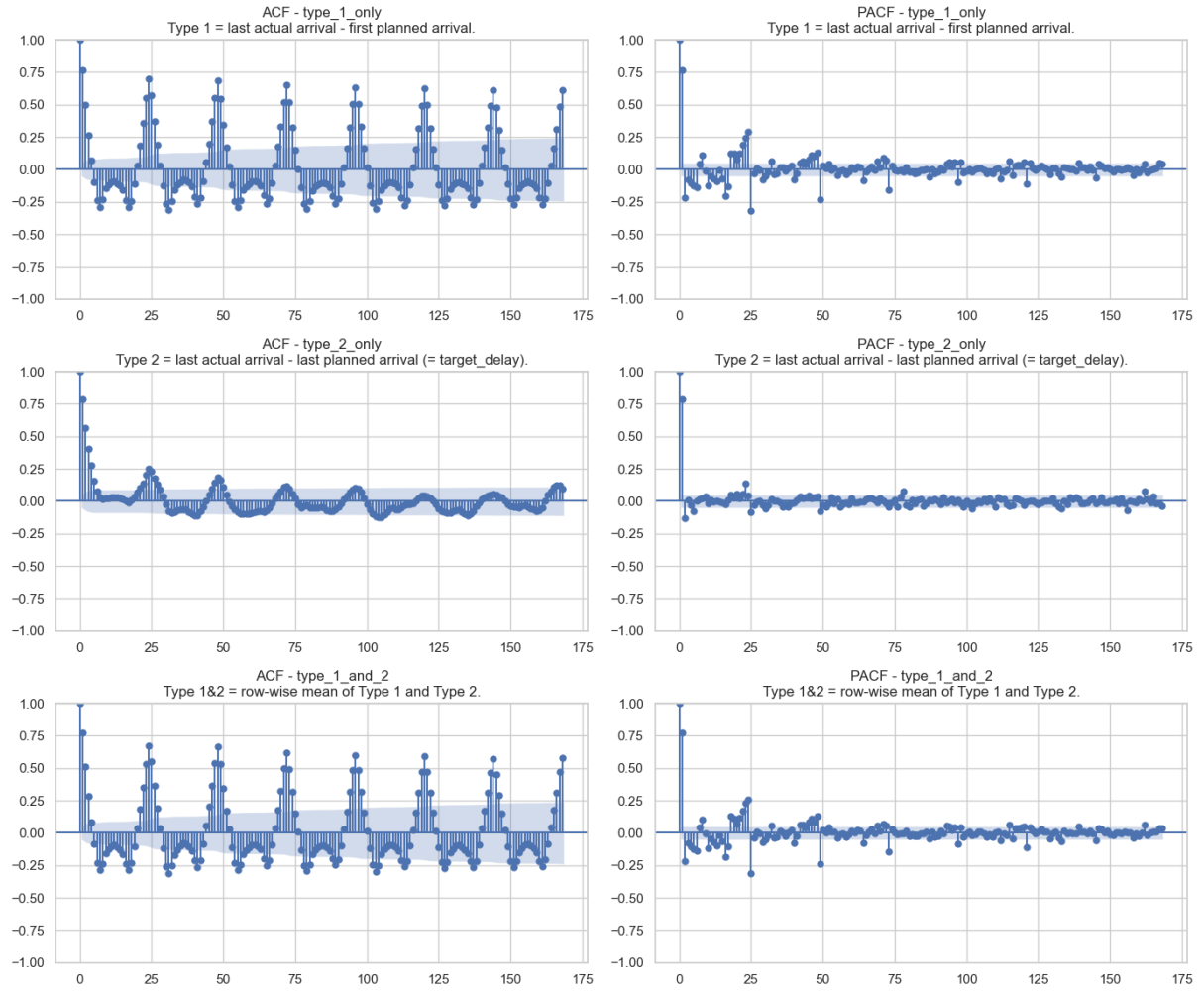


Figure 17: ACF (left) and PACF (right) up to lag 175 for all three target variants. ACF spikes at multiples of 24 confirm $s = 24$; PACF cut-off after lag 1 supports $p = 1$.

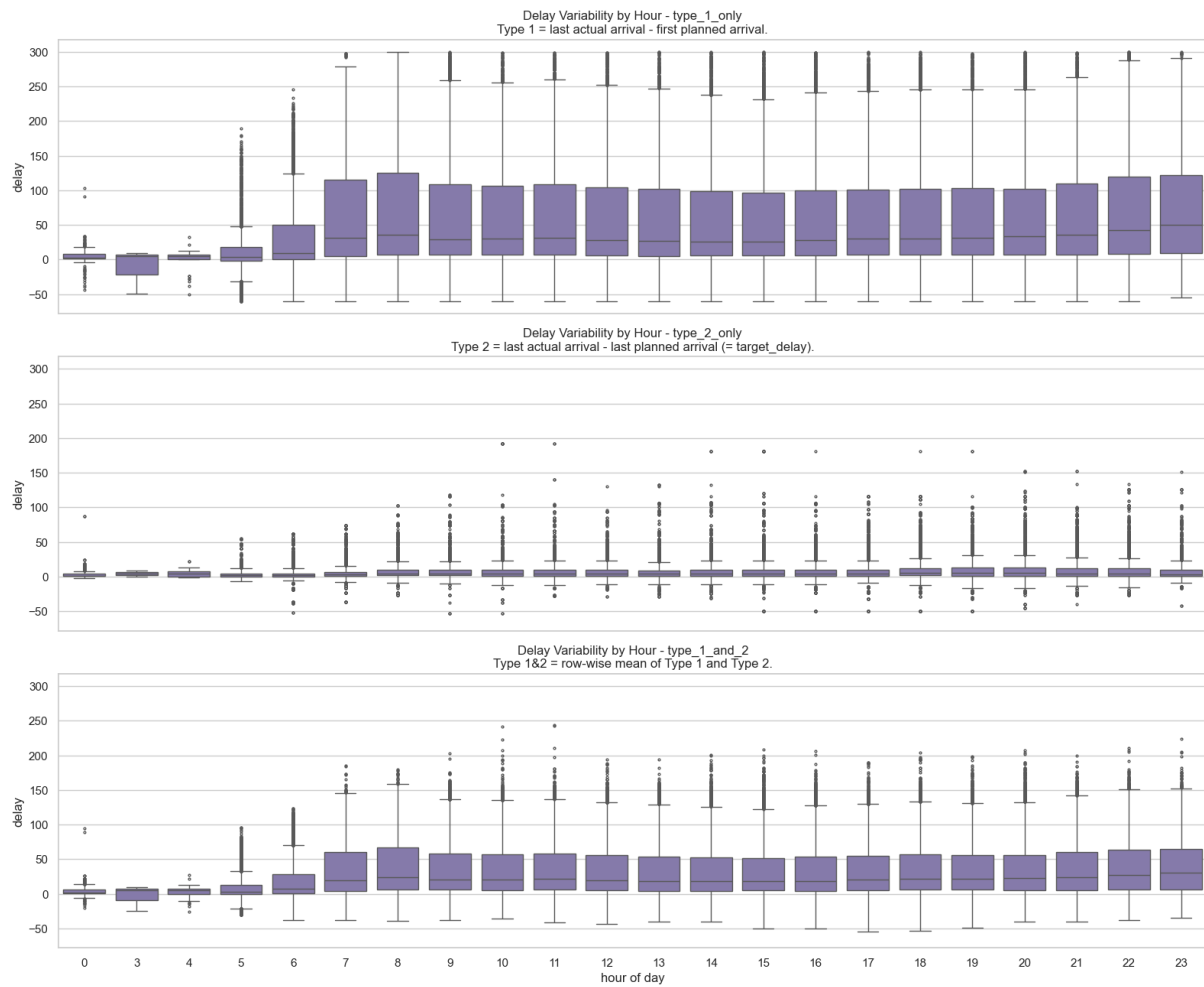


Figure 18: Delay variability by hour of day for all three target variants. Heteroscedasticity is visible in Type 2: variance peaks at 09:00–11:00 h and 20:00–22:00 h, motivating GARCH.

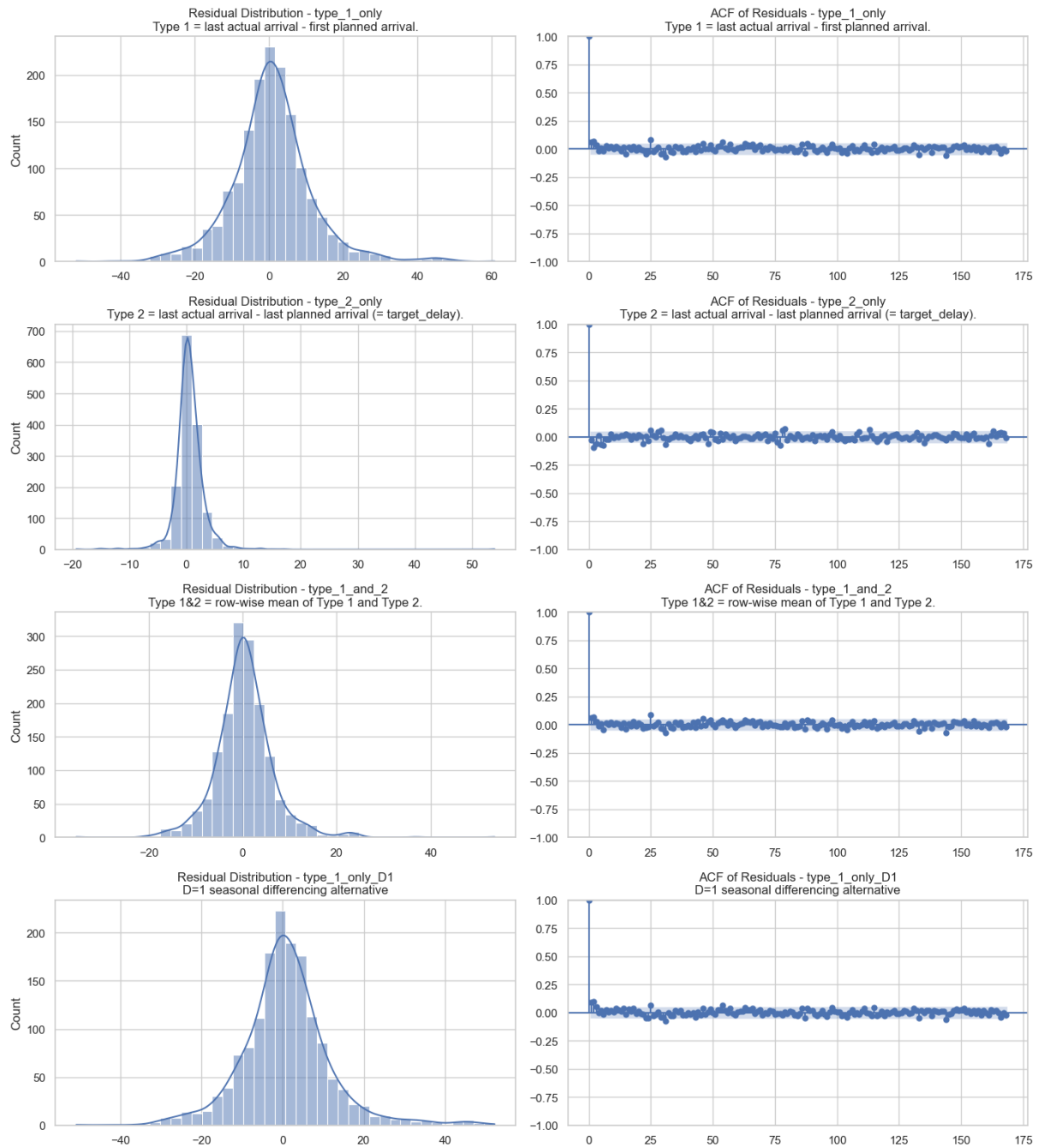


Figure 19: SARIMA residual diagnostics (4 rows). Rows 1–3: residual histogram and ACF of residuals (lags 0–175) for `type_1_only`, `type_2_only`, and `type_1_and_2`. Row 4: the $D = 1$ seasonal differencing alternative for `type_1_only`, included as a robustness check. For `type_2_only` (row 2) residuals are well within the 95% confidence bands and Ljung–Box confirms white noise at lags 24 ($p = 0.085$) and 168 ($p = 0.494$). For `type_1_only` (row 1) the ACF appears visually clean but Ljung–Box rejects white noise at both lags 24 and 168 ($p < 0.001$), indicating residual structure not captured by the univariate ARMA model. The $D = 1$ alternative (row 4) does not resolve this: Ljung–Box remains significant at both lags ($p < 0.001$), confirming that seasonal differencing is not the remedy.

C SARIMAX Supplementary Figures

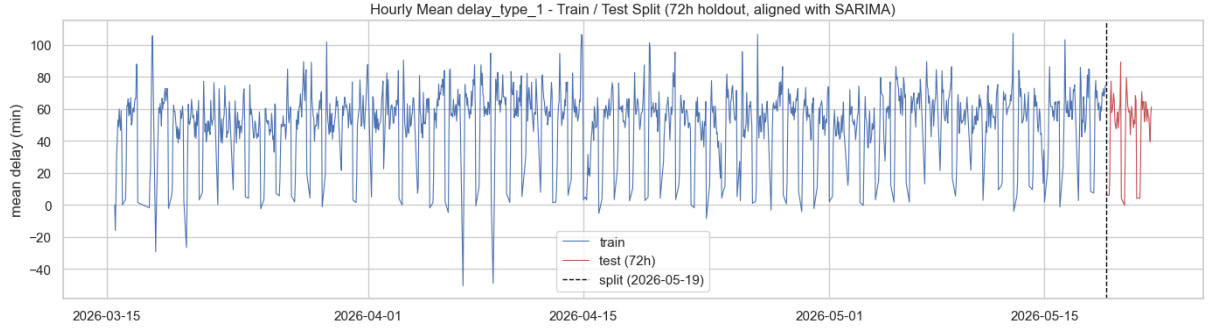


Figure 20: Train/test split for all three delay variants used in SARIMAX. `type_1_only` (blue) fluctuates around 50–75 min; `type_2_only` (red) is compressed to 0–15 min after Renfe adjustments; `type_1_and_2` (green) is their row-wise mean. The test window shows a calmer period than the April disruption spike.

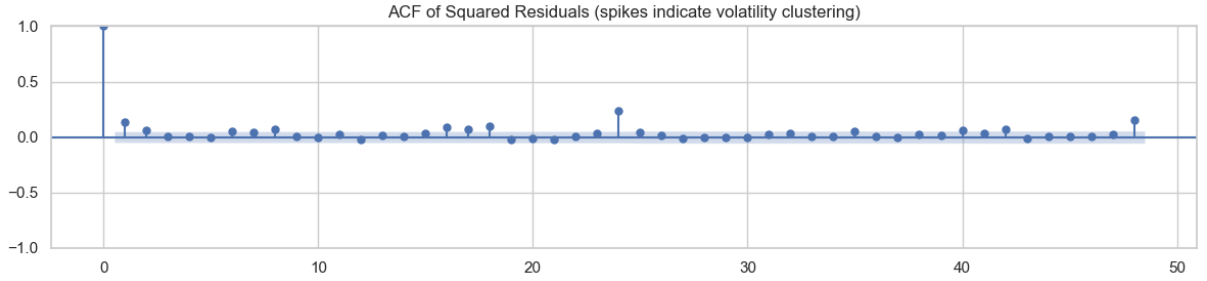


Figure 21: ACF of squared SARIMAX residuals (`type_1_only`, lags 0–50). All bars fall within the 95 % confidence band except for a clearly significant spike at lag 25 ($r \approx 0.30$), indicating residual volatility clustering at the daily seasonal frequency. The ARCH-LM test formally confirms this ($p < 0.001$): residual variance is not constant and spikes during disruption events. This motivates the GARCH(1, 1) model in Section 6.3, which captures this time-varying uncertainty explicitly.

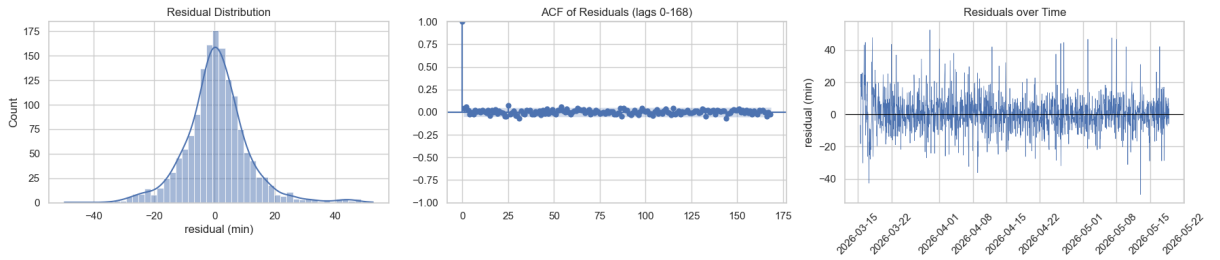


Figure 22: SARIMAX residual diagnostics for `type_1_only`: distribution (left), ACF of residuals (centre), residuals over time (right). The ACF is clean across all lags; residuals are centred at zero with spread ± 40 min reflecting the high variance of the unmasked series.

D GARCH Supplementary Figures

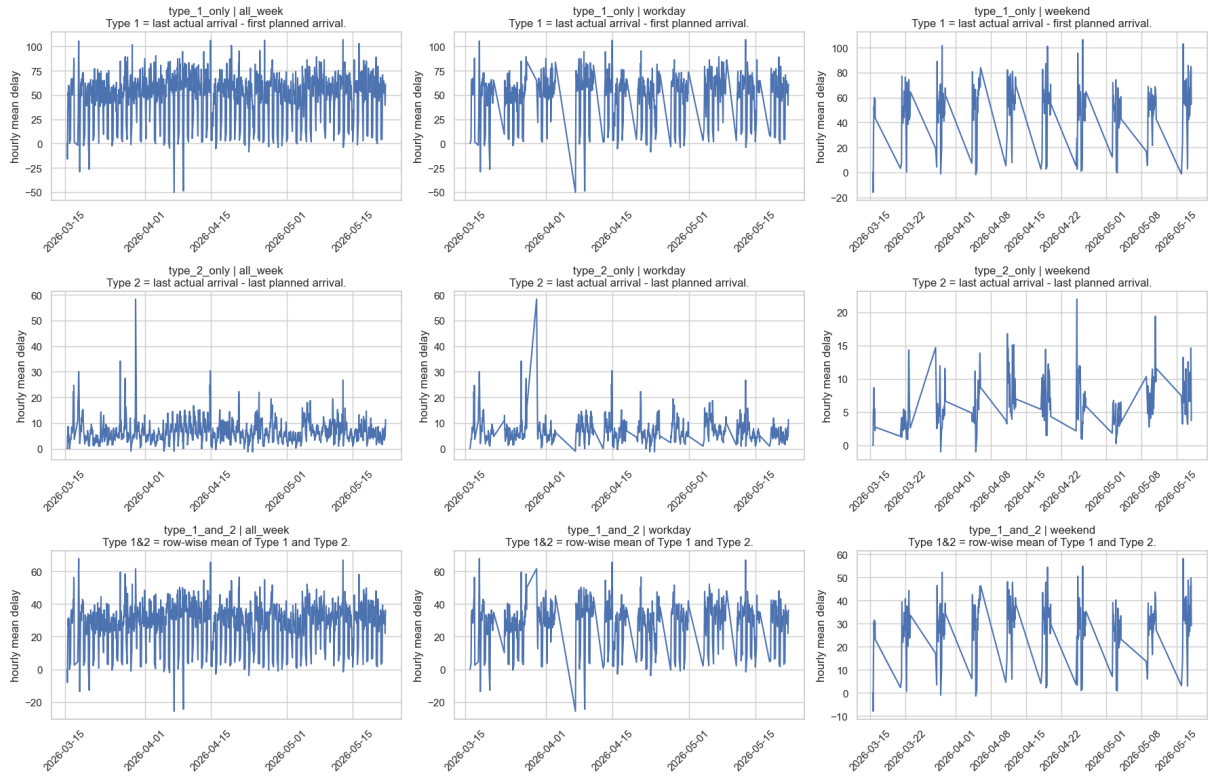
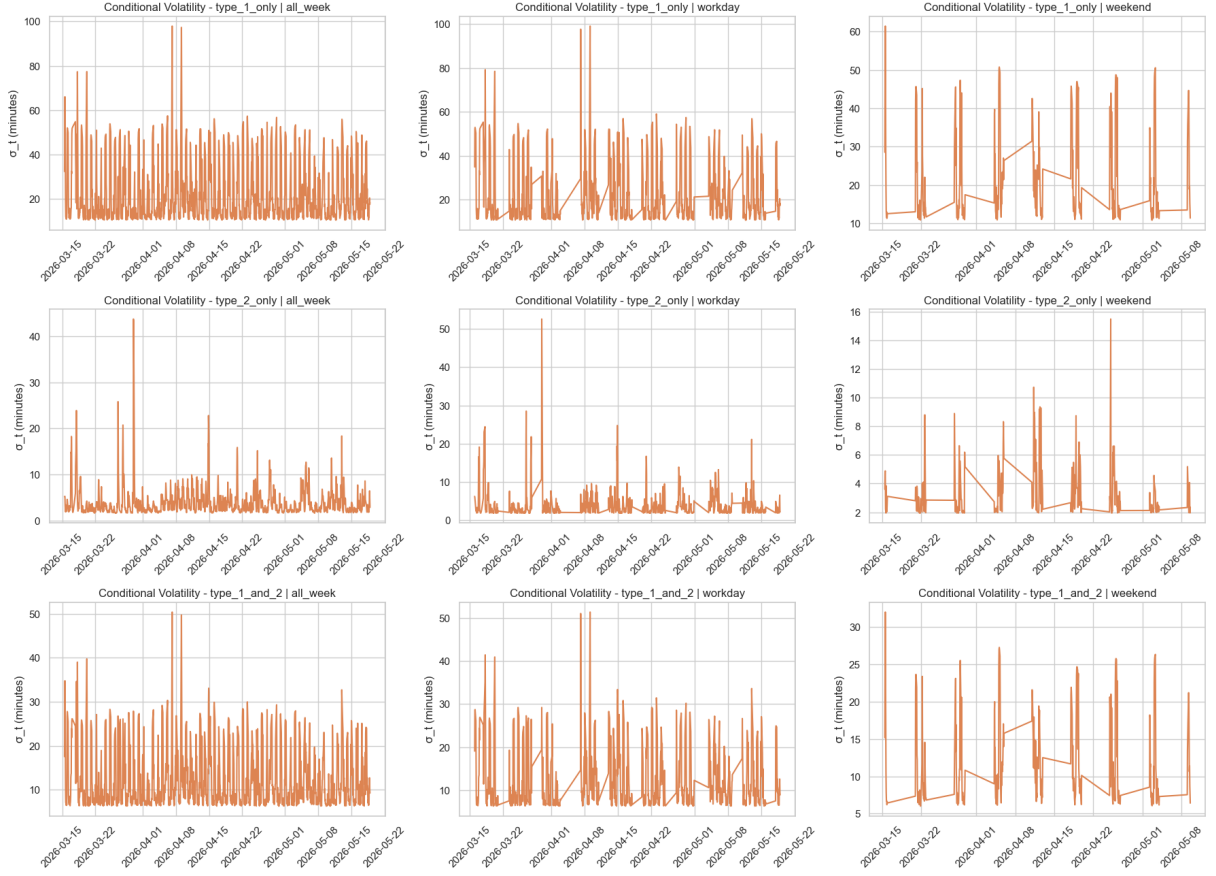


Figure 23: Hourly mean delay by delay variant and day-type segment used as GARCH input (3×3 grid). Visual heteroscedasticity is evident in all panels: the April disruption window shows both higher levels and wider swings.



References

- [1] *Rodalies de Catalunya*, Wikipedia, Accessed May 2026, 2026. [Online]. Available: https://en.wikipedia.org/wiki/Rodalies_de_Catalunya.
- [2] Agència Catalana de Notícies, “New delays and cancellations on rodalies commuter rail system due to strike,” *Catalan News*, Apr. 2025, Accessed: 2026-06-08. [Online]. Available: <https://www.catalannews.com/society-science/item/new-delays-and-cancellations-on-rodalies-commuter-rail-system-due-to-strike>.
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- [5] Europa Press, “Retrasos en la r1 y r4 de rodalies por incidencia entre l’hospitalet de llobregat y barcelona sants,” *Europa Press*, Apr. 2026, Accessed: 2026-06-08. [Online]. Available: <https://www.europapress.es/catalunya/noticia-retrasos-r1-r4-rodalies-incidencia-lhospitalet-llobregat-barcelona-sants-20260405120853.html>.